ATARS v2.0

Automated Time-Series Analysis and Reporting System

Comprehensive System Design & Architecture Report

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1. Executive Overview

ATARS v2.0 (Automated Time-Series Analysis and Reporting System) is a formally specified, domain-agnostic, end-to-end pipeline for automated statistical analysis, machine learning–enhanced anomaly detection, forecasting, and reproducible report generation from structured time-series environmental data.

The system was designed and implemented by Priyanshu, a second-year B.Tech (Computer Science) student at the Global Institute of Technology and Management, Haryana, India. It represents a novel contribution in the area of AI-grounded scientific reporting, introducing a Runtime Grounding Verifier (RGV) that programmatically enforces data provenance for every numerical claim made by a large language model.

Version 2.0 extends the v1 formal framework with two additive ML modules — Isolation Forest multivariate anomaly detection and Holt-Winters triple exponential smoothing forecasting — while preserving all 19 formal mathematical operators from the original theoretical specification.

System Name	ATARS — Automated Time-Series Analysis and Reporting System
Version	2.0.0
Author	Priyanshu
Institution	Global Institute of Technology and Management, Haryana, India
Domain	Environmental Air Quality (domain-agnostic design)
Input Format	CSV / Excel (hourly or sub-hourly sensor readings)
Output	Word (.docx) report, 15 publication-quality charts, JSON contract, audit log
Key Novelty	Runtime Grounding Verifier (RGV) — programmatic LLM hallucination detection
ML Additions (v2)	Isolation Forest (anomaly) + Holt-Winters (14-day PM10 forecast)
Security Model	All computation local. LLM receives aggregated JSON only. Zero telemetry.
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2. System Architecture & Pipeline

ATARS operates as a nine-step sequential pipeline. Each step has a clearly defined role, formal operator mapping, and auditable output. The pipeline is invoked from the command line and requires no manual intervention once launched.

2.1 Pipeline Overview (9 Steps)

Step	Name	Description
1	Data Ingestion	Load CSV/Excel, normalize columns, parse datetime, validate and sort. Handles multiple encodings (UTF-8, Latin-1, CP1252). Negative concentration values are coerced to NaN.
2	Quality Assessment	Assign quality flags $q_i \in \{0,1,2,3\}$ per observation. Compute daily Quality Score $Q(D) = N_v / N_{total}$. Flag days below minimum $Q(D) = 0.10$ threshold. Classify data confidence as HIGH / MODERATE / LOW.
3	Statistical Analysis	Compute daily aggregates (mean, median, std, IQR, p5, p95, N_v). Apply rolling 30-day baseline $B(D,W)$. Compute z-scores ζ_d, delta percentages δ_d, and 95% bootstrap confidence intervals for every pollutant and meteorological variable.
4	JSON Contract Assembly	Build the typed data contract J from all computed statistics. J encodes city metadata, overall Q(D), annual means, thresholds, exceedances, anomaly lists, OLS coefficients, correlation entries, and strict AI constraints preventing hallucination.
5	AI Narrative Generation	If LLM is enabled, send J to a local Ollama model (default: llama3.2) via HTTP to generate section narratives. Raw sensor data is never transmitted. Includes configurable timeout, retry, and fallback to statistical placeholders.
5.5	Runtime Grounding Verifier	The RGV (novel contribution) extracts every numerical value from every LLM sentence and verifies each against J using type-adaptive tolerance bounds. Computes G_{rate} = grounded / total numerical sentences. PASS threshold: $G_{rate} \geq 0.75$.
5.5b	Chart Insight Generator	Generates deterministic, data-driven textual insights for all 15 charts purely from computed values in daily_df. No LLM is involved. Eliminates hallucination risk for per-chart captions.
6	Chart Generation	Generate 15 publication-quality Matplotlib/Seaborn figures at 160 DPI. Charts cover time-series, distributions, z-score heatmaps, correlation matrices, ACF/PACF, diurnal patterns, VOC panels, calendar heatmaps, ML anomaly comparison, and Holt-Winters forecast.
7	Report Compilation	Assemble the full Word (.docx) report using python-docx. Includes cover page, TOC, executive summary, all 15 charts with insights, statistical tables, OLS coefficients, correlation matrix, ML results, RGV grounding report, and methodology appendix.

8	Audit Log	Write a reproducibility audit log containing: run ID (UUID), Hash(J) (SHA-256 of JSON contract), pipeline timestamp, data file info, all configuration parameters, ML results summary, and grounding verification verdict.

2.2 Command-Line Interface

ATARS is invoked from the terminal. Core flags:

```
python atars.py --data your_data.csv --city "Delhi"
python atars.py --data your_data.csv --city "Delhi" --no-llm
python atars.py --data your_data.csv --city "Delhi" --no-llm --no-forecast
```

Flag	Purpose
--data	Path to input CSV or Excel file
--city	Location name for report title and metadata
--station	Station identifier (default: Station-01)
--output	Output directory (default: atars_output)
--no-llm	Disable LLM narrative; use statistical placeholder text
--llm-model	Ollama model name (e.g. llama3.2, mistral, phi3)
--baseline	Rolling baseline window W in days (default: 30)
--timeout	LLM timeout per section in seconds (default: 600)
--retry	Number of LLM retry attempts per section (default: 2)
--no-forecast	Disable Holt-Winters PM10 forecast (Chart 15)

3. Data Layer — Ingestion & Preprocessing

3.1 Supported Input Formats

ATARS accepts CSV and Excel files. The ingestion module tries four encodings in sequence — UTF-8, Latin-1, CP1252, and UTF-8-BOM — to handle real-world file encoding issues. Column names are normalized via a keyword-matching dictionary that maps raw sensor column names (potentially containing encoding artifacts such as Â° or Âµg) to clean internal names.

3.2 Supported Variables

Variable	Internal Name	Reference Threshold
PM10	PM10	45.0 µg/m³ (WHO AQG 2021, 24h mean)
Nitrogen Dioxide	NO2	25.0 µg/m³ (WHO AQG 2021)
Sulphur Dioxide	SO2	40.0 µg/m³ (WHO AQG 2021)
Carbon Monoxide	CO	4.0 mg/m³ (WHO AQG 2021)
Ozone	Ozone	100.0 µg/m³ (WHO AQG 2021, peak season)
Ammonia	NH3	400.0 µg/m³ (CPCB India standard)
Nitric Oxide	NO	No WHO guideline — informational
Nitrogen Oxides	NOx	No WHO guideline — informational
Benzene	Benzene	1.7 µg/m³ (WHO annual reference, IARC Group 1)
Toluene / Xylene / Eth-Benzene / MP-Xylene	Toluene / Xylene / Eth_Benzene / MP_Xylene	No WHO guideline — informational
Humidity	humidity_pct	Meteorological (0–100%)
Rainfall	rain_mm	Meteorological (mm)
Atmospheric Pressure	pressure_hpa	Meteorological (hPa)
Wind Speed 10m / 100m	wind_speed_10m / wind_speed_100m	Meteorological (m/s)
Wind Direction 10m / 100m	wind_dir_10m / wind_dir_100m	Meteorological (degrees)

3.3 Data Validation Rules

Every loaded column is subjected to the following validation chain before entering the pipeline:

- Negative concentration values are physically impossible and are set to NaN.
- Each variable is range-checked against physical plausibility bounds (e.g., PM10: 0–1000 µg/m³, pressure: 800–1100 hPa).
- Values outside 5×IQR are flagged as suspect ($q_i = 2$) rather than immediately discarded.
- Missing or out-of-range values are flagged as invalid ($q_i = 3$).
- Duplicate columns are removed (first occurrence kept).

4. Formal Mathematical Framework

ATARS implements 19 formal mathematical operators derived from the ATARS theoretical framework. These operators are the foundational building blocks for all statistical analysis in the system and are preserved unchanged in v2.0.

Equation	Operator	Description
Eq. 1	$D = \{(t_i, x_i, q_i)\}$	Observation space — each record is a (timestamp, value, quality flag) triple.
Eq. 2	$Q(D) = N_v / N_{total}$	Data Quality Score — ratio of valid observations to total observations.
Eq. 3	$q_i \in \{0, 1, 2, 3\}$	Quality flag: 0=unreviewed, 1=valid, 2=suspect, 3=invalid.
Eq. 4	$A(D_v) = \text{aggregate}(x_i)$	Aggregation operator — computes mean, median, std, IQR, p5, p95, N_v for valid observations.
Eq. 5	$B(D, W)$	Rolling 30-day baseline — sliding window mean and standard deviation.
Eq. 6	$CI(\alpha)$	Bootstrap 95% confidence interval (B=1000 samples) for daily means.
Eq. 7	$\zeta_d = (\mu_d - \mu_W) / \sigma_W$	Z-score anomaly operator — normalized deviation from baseline.
Eq. 8	$\delta_d = (\mu_d - \mu_{d-1}) / \mu_{d-1}$	Day-over-day delta percentage — short-term trend indicator.
Eq. 9	$\lvert\zeta_d\rvert > 3.0 \rightarrow \text{anomaly}$	Anomaly detection condition — z-score threshold at 3σ.
Eq. 10	$\text{exceeds_thr}: \mu_d > \text{threshold}$	WHO/CPCB exceedance flag per variable per day.
Eq. 11	$Q(D)$ classified	Confidence classification: ≥0.90 = HIGH, ≥0.70 = MODERATE, else LOW.
Eq. 12	$\hat{\rho}(k) = \text{Corr}(x_t, x_{t-k})$	Sample ACF — autocorrelation function up to 48 lags, manual numpy implementation.
Eq. 13	$\hat{\varphi}_{kk}$ (Yule-Walker)	Sample PACF — partial autocorrelation via OLS recursion, manual implementation.
Eq. 14	$\hat{\beta} = (X^T X)^{-1} X^T y$	OLS regression — predicts PM10 from meteorological predictors.
Eq. 15	$R^2 = 1 - SS_{res} / SS_{tot}$	Coefficient of determination for OLS fit quality.

Eq. 16	$X_s = \text{StandardScaler}(X)$	Feature standardisation before OLS (zero-mean, unit-variance).
Eq. 17	$\text{Corr}(X,Y) \not\equiv C(X \to Y)$	Association caveat — correlation does not imply causation (enforced in report text).
Eq. 18	$R_{jk} = \text{Corr}(X_j, X_k)$	Pearson correlation matrix — all pollutant and meteorological cross-correlations.
Eq. 19	$G(s) \subseteq J \;\; \forall s \in N$	Grounding constraint — every numerical value in every LLM sentence must be traceable to J.

5. Quality Assessment Engine

The Quality Assessment Engine is one of the most distinctive components of ATARS. Rather than silently discarding or imputing bad data, it assigns an explicit quality flag to every observation and propagates quality metadata through all downstream computations.

5.1 Quality Flag System

Every observation record receives a quality flag q_i assigned by the following decision logic, evaluated in order:

Flag	Label	Condition
0	Unreviewed	Default for non-pollutant / meteorological columns
1	Valid	Not NaN, within physical bounds, within 5×IQR
2	Suspect	Not NaN, within physical bounds, but outside 5×IQR statistical fence
3	Invalid	NaN, negative, or outside physical plausibility bounds

5.2 Daily Quality Score Q(D)

For each day and each variable, ATARS computes $Q(D) = N_v / N_{total}$, where N_v is the count of valid (flag=1) observations and N_{total} is the count of all observations. Days where $Q(D) < 0.10$ (configurable via min_valid_ratio) are excluded from statistical analysis. This ensures that statistics are never computed on fewer than 10% valid readings.

5.3 Confidence Classification

The daily Q(D) scores drive a confidence flag that propagates into the JSON contract and ultimately into the generated report:

- Q(D) ≥ 0.90 → HIGH confidence — data is dense and reliable.
- 0.70 ≤ Q(D) < 0.90 → MODERATE confidence — minor gaps present.
- Q(D) < 0.70 → LOW confidence — significant missing data; statistics carry higher uncertainty.

6. Statistical Analysis Engine

6.1 Daily Aggregation

After quality filtering, ATARS aggregates hourly (or sub-hourly) observations to daily statistics. For each variable on each day, the following quantities are computed: arithmetic mean, median, standard deviation, inter-quartile range (IQR), 5th and 95th percentiles, and count of valid observations N_v.

6.2 Rolling Baseline & Anomaly Detection

A 30-day (configurable) rolling window baseline B(D,W) is computed for each variable. The baseline provides a dynamic reference μ_W and σ_W against which each daily mean is assessed. The z-score $\zeta_d = (\mu_d - \mu_W) / \sigma_W$ measures the normalized deviation from this baseline. Days where $|\zeta_d| > 3.0$ are flagged as statistical anomalies (Equation 9). The day-over-day delta δ_d tracks short-term rate of change.

6.3 Confidence Intervals

Bootstrap confidence intervals (B = 1000 resamples, α = 0.05) are computed for each daily mean. This non-parametric approach makes no distributional assumption and is robust to the skewed concentration distributions typical in air quality data. The 95% CI is recorded in the JSON contract and displayed as shaded bands in the time-series charts.

6.4 Temporal Analysis — ACF & PACF

For PM10 (and other available variables), the system computes the Sample Autocorrelation Function (ACF) up to 48 lags and the Sample Partial Autocorrelation Function (PACF) up to 24 lags. Both are implemented manually using pure NumPy — no external libraries such as statsmodels are required. The 95% confidence bound ($\pm 1.96/\sqrt{n}$) is plotted alongside the ACF and PACF to identify significant periodicity.

6.5 Correlation Matrix

A full Pearson correlation matrix R is computed across all available pollutant and meteorological variables using daily means. The result is a d×d matrix (where d = number of available variables) and is visualized as a color-coded heatmap in Chart 6. Critically, the system enforces the association-not-causation caveat (Equation 17) in the report narrative.

6.6 OLS Regression

Ordinary Least Squares (OLS) regression is run with PM10 as the response variable and meteorological predictors (humidity, wind speed, rainfall, atmospheric pressure) as explanatory variables. Features are standardized before fitting. The system reports R^2, the regression coefficients $\hat{\beta}$ for each predictor, the intercept, and the number of observations. As with the correlation matrix, the report explicitly states that this represents statistical association only, not a causal mechanism.

7. Machine Learning Layer (v2.0 Additions)

ATARS v2.0 introduces two ML modules as additive enhancements to the formal statistical framework. Neither module replaces any formal operator — they extend the system's detection and forecasting capabilities while preserving full reproducibility.

7.1 ML-1: Isolation Forest Multivariate Anomaly Detection

The Isolation Forest (IF) module addresses the primary limitation of z-score anomaly detection: its univariate nature. The z-score operator checks one variable at a time, meaning it can miss days where no single pollutant exceeds 3σ but the joint combination of PM10 + NO2 + SO2 + CO + Ozone is abnormal.

7.1.1 Algorithm Details

- Features: PM10, NO2, SO2, CO, Ozone (configurable via if_cols).
- Approach: Builds a wide matrix (rows = days, columns = pollutants), drops incomplete rows.
- Preprocessing: StandardScaler (zero mean, unit variance) — required as IF is not scale-invariant.
- Model: sklearn IsolationForest with n_estimators=150, contamination=0.05, random_state=42.
- Output: Each day is scored and predicted as normal (+1) or anomaly (−1).
- Minimum data: At least 30 complete rows with 2+ feature columns required.

7.1.2 Comparison with Z-Score

After running both methods, ATARS classifies each anomaly day into one of four categories:

- Both: Day flagged as anomaly by both IF and z-score — high-confidence anomaly.

- IF-only: Day flagged by IF but not z-score — multivariate pattern, no univariate spike.
- Z-score only: Day flagged by z-score but not IF — single-variable spike within multivariate norm.
- Neither: Normal day according to both methods.

IF-only days represent the genuine added diagnostic value of the v2.0 ML layer — these are anomalies that the formal ζ_d operator would have missed entirely.

7.1.3 Reproducibility & Security

fixed random_state=42 ensures that re-running on identical data produces identical anomaly flags. All computation occurs in memory on the local machine. No data is transmitted externally.

7.2 ML-2: Holt-Winters Triple Exponential Smoothing Forecast

The Holt-Winters (HW) module provides a 14-day ahead forecast of daily PM10 concentrations. It is implemented as a pure NumPy function with zero additional dependencies beyond the existing stack.

7.2.1 Model Structure

ATARS uses the additive Holt-Winters model with three smoothing equations:

- Level equation: $L_t = \alpha(x_t - S_{t-m}) + (1-\alpha)(L_{t-1} + T_{t-1})$
- Trend equation: $T_t = \beta(L_t - L_{t-1}) + (1-\beta)T_{t-1}$
- Seasonal equation: $S_t = \gamma(x_t - L_t) + (1-\gamma)S_{t-m}$
- Forecast: $\hat{x}_{t+h} = L_t + h \cdot T_t + S_{t-m+((h-1) \bmod m)+1}$

Default parameters: $\alpha=0.3$ (level), $\beta=0.1$ (trend), $\gamma=0.2$ (seasonal), seasonal period $m=7$ days (weekly). These are all configurable in DEFAULT_CONFIG.

7.2.2 Prediction Intervals

80% prediction intervals are computed as $\pm 1.28 \times \sigma_{residual} \times \sqrt{h}$, where h is the forecast horizon. Intervals widen with increasing horizon to reflect growing uncertainty. All forecast values are clipped to ≥ 0 (concentrations cannot be negative).

7.2.3 Security Design Decision

Forecast values are deliberately NOT included in the JSON contract J. If forecast values were in J, the LLM could cite them as historical facts in the narrative — creating a factual inaccuracy. Instead, forecast results appear only in Chart 15 and the dedicated ML section of the report. The system prints an explicit security note explaining this design choice.

7.2.4 Error Metrics

The model reports in-sample RMSE (Root Mean Squared Error) and MAE (Mean Absolute Error) computed against the full historical PM10 series. These metrics characterize how well the smoothed model fits the observed data.

8. Runtime Grounding Verifier (RGV) — Novel Contribution

The Runtime Grounding Verifier (RGV) is the primary novel scientific contribution of the ATARS project. It addresses a fundamental challenge in AI-assisted scientific reporting: large language models can generate plausible-sounding numerical claims that are not actually grounded in the input data (hallucination).

8.1 Problem Statement

When an LLM generates narrative text from a JSON data contract, each sentence may contain numbers. A number in the narrative should match a corresponding value in J within an acceptable tolerance. Without systematic checking, it is impossible to know whether the LLM has faithfully reported the data or invented values.

8.2 RGV Algorithm (Equation 19: $G(s) \subseteq J \ \forall \ s \in N$)

The RGV implements the following verification algorithm:

- Step 1 — JSON Lookup Table: Build a comprehensive lookup dictionary from J containing every numerical value across all fields, with their labels and associated types.
- Step 2 — Sentence Segmentation: Split the full LLM narrative N into individual sentences s.
- Step 3 — Number Extraction: Use regex to extract all numeric tokens from each sentence s.
- Step 4 — Tolerance-Adaptive Matching: For each extracted number, search the J lookup using type-specific tolerance:

Value Type	Tolerance	Example
Concentrations / means	±5%	PM10 mean: 67.3 → matches J if 64.0–70.7
Count / integer fields	±2	Anomaly count: 18 → matches J if 16–20
Quality scores (Q, R^2)	±0.1	Q(D) = 0.87 → matches J if 0.77–0.97
Percentages, thresholds	±5%	WHO 45 μg/m³ → matches J if 42.75–47.25

- Step 5 — Sentence Classification: Each sentence is classified as GROUNDED (all numbers match J), PARTIALLY (≥50% match), UNGROUNDED (<50% match), or NON_NUMERICAL (no numbers — excluded from metric).
- Step 6 — G_rate Computation: G_rate = grounded_sentences / total_numerical_sentences.

- Step 7 — Verdict: PASS if G_rate ≥ 0.75, WARNING if 0.50 ≤ G_rate < 0.75, FAIL if G_rate < 0.50.

8.3 Report Integration

Every sentence flagged as UNGROUNDED is highlighted in the Word report with a warning annotation. Chart 13 visualizes the full grounding status breakdown (grounded / partially / ungrounded / non-numerical) as a stacked bar chart for each report section. The grounding verification results are also saved to grounding_verification.json for independent audit.

8.4 Why This Matters

The RGV transforms AI-assisted reporting from a black-box process into an auditable, verifiable one. Reviewers, regulators, and journal editors can inspect the G_rate and the ungrounded sentence log to understand exactly which claims are backed by the data contract and which may require additional scrutiny. This is particularly critical in environmental and regulatory reporting contexts.

9. Chart Generation — 15 Publication-Quality Figures

ATARS generates 15 figures using Matplotlib and Seaborn at 160 DPI, saved as high-resolution PNG files. Every chart is accompanied by a deterministic data-driven insight caption generated by the Chart Insight Generator (Step 5.5b), which computes text directly from daily_df values without LLM involvement.

Chart	Title	Content
1	Time-Series Overview	Hourly PM10 and NO2 with daily mean overlay, WHO guideline lines, and exceedance zones shaded in red.
2	Distribution Panel	Kernel density + violin plots for all major pollutants showing annual concentration distributions.
3	Z-Score Heatmap	Daily z-score ζ_d heatmap for all pollutants over time. Anomaly days (|ζ| > 3) highlighted in deep red/blue.
4	Seasonal Monthly Box Plots	Monthly box plots for PM10 and NO2 showing seasonal variation, median, IQR, and WHO threshold.
5	Data Quality Dashboard	Daily Q(D) heatmap by month, overall quality distribution histogram, and variable-level completeness bar chart.
6	Correlation Heatmap	Full Pearson correlation matrix for all pollutants and meteorological variables with annotated r values.

7	OLS Regression Panel	Scatter plots and regression lines for PM10 vs. each meteorological predictor with R^2 annotation.
8	Diurnal Pattern	Hour-of-day average concentration profiles for PM10, NO2, and CO showing rush-hour and nocturnal peaks.
9	Day-of-Week Pattern	Average PM10 concentration by day of week showing weekday vs. weekend contrast.
10	VOC / BTEX Panel	Time-series and bar charts for Benzene, Toluene, Ethylbenzene, and Xylene with WHO annual reference lines.
11	Rain vs. Pollution Scatter	Scatter plots of daily rainfall against PM10 and SO2 with OLS trend lines (wet deposition association).
12	Exceedance Calendar Heatmap	Full-year calendar heatmap of daily PM10 means. WHO exceedance days marked with ×.
13	RGV Grounding Visualization	Stacked bar charts showing grounding status per report section — proportion GROUNDED / PARTIALLY / UNGROUNDED / NON_NUMERICAL.
14	ML Anomaly Comparison	Timeline showing IF anomaly score, z-score, and the four agreement categories (both / IF-only / z-only / neither).
15	Holt-Winters PM10 Forecast	14-day PM10 forecast with 80% prediction interval bands, historical series, and WHO guideline overlay.

10. JSON Data Contract & Security Model

10.1 The JSON Contract J

The JSON contract J is the single authoritative data object that separates the statistical computation layer from the AI narrative generation layer. It is assembled in Step 4 from all computed statistics and serves as the sole input to the LLM.

J contains the following information classes:

- City and station metadata (name, date range, reporting period).
- Overall data quality Q(D) with confidence classification.
- Annual means, maxima, minima, standard deviations for all variables.
- WHO/CPCB threshold values and exceedance day counts per variable.
- Anomaly day lists (dates + z-scores) per variable.
- OLS regression coefficients and R^2.
- Correlation matrix entries (top correlations only, to reduce token count).
- AI constraints — explicit instructions in J that forbid the LLM from extrapolating, speculating, or inventing values not present in J.

10.2 Security Properties

The ATARS security model enforces the following properties:

- Data minimization: The LLM receives J (aggregated statistics) — never the raw hourly sensor data. This limits the LLM's ability to reason about individual readings and reduces exposure.
- Local computation: All statistical and ML computation runs entirely in memory on the host machine. No computation is offloaded to external services.
- Offline LLM: The system uses Ollama to run the LLM locally. Zero cloud API calls, zero telemetry, zero data leaving the machine.
- Reproducibility: Isolation Forest uses fixed random_state=42. The SHA-256 hash of J is stored in the audit log, enabling independent verification that the same data produces the same report.
- Forecast isolation: Holt-Winters forecast values are deliberately excluded from J to prevent the LLM from citing forecasts as historical facts.

11. Report Compilation & Audit Log

11.1 Word Report Structure

The Word (.docx) report is assembled in Step 7 using python-docx. The report follows the structure of a formal scientific report and contains the following sections:

- Cover page with title, author, institution, station, and date range.
- Table of Contents.
- Executive Summary with key metrics and overall air quality verdict.
- Section 1: Data Overview — variable coverage, date range, Q(D) summary.
- Section 2: Statistical Analysis — all 15 charts embedded with captions and insight text.
- Section 3: Anomaly & Exceedance Analysis — anomaly tables, exceedance counts.
- Section 4: Correlation & Regression — matrix visualization and OLS results.
- Section 5: ML Results — Isolation Forest comparison and Holt-Winters forecast.
- Section 6: AI Narrative — LLM-generated text with grounding annotations.
- Section 7: RGV Grounding Report — per-sentence grounding status table.
- Appendix: Methodology, formal operators, configuration parameters used.

11.2 Audit Log

The audit log (audit_log.json) written in Step 8 contains:

- run_id: UUID uniquely identifying this pipeline execution.
- hash_J: SHA-256 hex digest of the JSON contract J for reproducibility verification.
- timestamp: ISO 8601 datetime of the pipeline run.
- data_file, data_shape, date_range: Input data provenance.

- config_snapshot: Full copy of all configuration parameters used.
- ml_summary: IF anomaly counts and HW RMSE/MAE metrics.
- grounding_verdict: Overall G_rate and PASS/WARNING/FAIL verdict.

The audit log enables exact reproduction of any prior report by re-running with the same data file and config, and verifying that hash_J matches.

12. System Configuration

All configurable parameters are defined in a single DEFAULT_CONFIG dictionary at the top of the source file. This design ensures that domain adaptation requires editing only one location.

Parameter	Default	Description
baseline_window_days	30	Rolling window W for baseline mean/std computation.
anomaly_z_threshold	3.0	Z-score boundary for statistical anomaly flagging.
confidence_high	0.90	Q(D) threshold for HIGH confidence classification.
confidence_moderate	0.70	Q(D) threshold for MODERATE confidence classification.
ci_alpha	0.05	Alpha level for 95% bootstrap confidence intervals.
acf_lags	48	Number of lags computed for ACF/PACF analysis.
bootstrap_samples	1000	Number of bootstrap resamples for CI computation.
min_valid_ratio	0.10	Minimum Q(D) to include a day in analysis.
if_contamination	0.05	Expected anomaly fraction for Isolation Forest (5%).
if_n_estimators	150	Number of trees in the Isolation Forest ensemble.
if_random_state	42	Fixed seed for reproducible IF results.
hw_forecast_days	14	Number of days ahead for Holt-Winters PM10 forecast.
hw_seasonal_periods	7	Seasonal period for HW model (7 = weekly).
hw_alpha / beta / gamma	0.3 / 0.1 / 0.2	Holt-Winters smoothing parameters for level, trend, seasonal.
llm_model	llama3.2	Ollama model name for narrative generation.
llm_timeout	600	Timeout per LLM section in seconds.

llm_retry	2	Number of retry attempts on LLM timeout.

13. Dependencies & Deployment

13.1 Python Dependencies

```
pip install pandas numpy scipy matplotlib seaborn python-docx scikit-learn
```

Package	Role in ATARS
pandas	Data ingestion, column normalization, datetime parsing, pivot tables for daily aggregation and correlation.
numpy	All numerical computation — ACF/PACF, z-scores, bootstrap resampling, Holt-Winters smoothing.
scipy	Statistical distributions for confidence bound computation; supplementary stats utilities.
matplotlib	All 15 publication-quality charts at 160 DPI. Custom color palette applied via rcParams.
seaborn	Correlation heatmaps, distribution plots, KDE overlays.
python-docx	Word (.docx) report assembly — text, tables, embedded images.
scikit-learn	IsolationForest (ML-1 anomaly detection) and StandardScaler (feature normalization).

13.2 Optional — LLM Narrative

The LLM narrative layer requires Ollama (https://ollama.com), a free, offline, locally-hosted LLM runtime. The recommended model is llama3.2 (3B or 8B). Other compatible models include mistral and phi3. If LLM is unavailable or the --no-llm flag is passed, all narrative sections fall back to deterministic statistical placeholder text, and the full pipeline completes without any LLM dependency.

13.3 Automation — Windows Task Scheduler

ATARS automatically generates a Windows batch file (run_atars.bat) in the output directory at the end of each run. This file can be registered with Windows Task Scheduler to run ATARS daily at a scheduled time, enabling fully automated daily air quality reporting as long as the input CSV file is updated before the scheduled execution time.

14. Version History — v1 to v2 Changes

ATARS v2.0 is a strictly additive extension of v1. All formal operators (Equations 1–19) and the JSON contract security model are preserved unchanged.

Component	v2.0 Change
Formal operators (Eq. 1–18)	Unchanged — all preserved exactly as in v1.
JSON contract J	Unchanged — structure and security model preserved. Forecast values explicitly excluded.
ML-1: Isolation Forest	NEW in v2.0 — multivariate anomaly detection. Runs alongside z-score (does not replace it).
ML-2: Holt-Winters Forecast	NEW in v2.0 — 14-day PM10 forecast (Chart 15). Implemented in pure NumPy. Excluded from J.
Chart Insight Generator	NEW in v2.0 — deterministic LLM-free captions for all 15 charts. All numbers from daily_df.
Equation 19 (RGV)	NEW in v2.0 — $G(s) \subseteq J$ formal grounding constraint. Implemented as the RuntimeGroundingVerifier class.
Chart 13 (RGV Visualization)	NEW in v2.0 — stacked bar chart of grounding status per report section.
Chart 14 (ML Anomaly)	NEW in v2.0 — Isolation Forest score timeline and agreement comparison with z-score.
Chart 15 (Forecast)	NEW in v2.0 — Holt-Winters 14-day PM10 forecast with 80% prediction intervals.

15. Known Limitations & Future Directions

15.1 Current Limitations

- Holt-Winters smoothing parameters (α, β, γ) are user-set constants rather than optimized via cross-validation. This may reduce forecast accuracy for datasets with unusual seasonality.
- The OLS regression is linear and does not capture non-linear meteorological interactions (e.g., wind direction × speed combined effects on dispersion).

- Isolation Forest contamination parameter is fixed at 5%, meaning the model always flags approximately 5% of days as anomalies regardless of true prevalence.
- The RGV tolerance bounds (±5%, ±2, ±0.1) are fixed heuristics. Domain-specific datasets may require adjustment.
- Chart 12 (exceedance calendar) is optimized for PM10. A full multi-pollutant calendar is not yet implemented.
- The Ollama LLM is single-model — the system cannot run an ensemble of models and cross-validate narrative consistency.

15.2 Future Directions

- ARIMA/SARIMA modeling with auto-hyperparameter selection for improved forecasting accuracy.
- Automated cross-validation for Holt-Winters smoothing parameter optimization.
- Extension to multi-station networks with spatial interpolation and cross-station anomaly correlation.
- Integration of a second LLM for RGV-style narrative consistency checking between two independently generated reports.
- Web dashboard interface (Streamlit or Dash) for interactive exploration of ATARS outputs.
- CPCB AQI computation and AQI category classification as an additional output layer.

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